

Sonali Godavarthy

AI Engineer

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📍 Siegen, Germany (Open to Relocate) [in](#) LinkedIn [G](#)itHub [G](#)oogle Scholar



PROFILE

AI Engineer specializing in generative AI, computer vision, and applied ML, with production experience spanning diffusion models, retrieval-augmented LLM systems, and vision-language pipelines. Builds and scales training/evaluation workflows on SLURM-based HPC clusters, backed by a systems-engineering foundation in cloud infrastructure and observability. Focused on moving models from experimentation into deployable, production-grade systems.

EDUCATION

M.Sc. Computer Science (Visual Computing), University of Siegen 10/2023 – 07/2026 | Siegen, Germany
GPA: 1.6 (German scale)

B.E. Information Technology, Vasavi College of Engineering 07/2018 – 06/2022 | Hyderabad, India
GPA: 1.52 (German scale)

SKILLS

Generative AI & Diffusion Models Stable Diffusion, FLUX, Diffusers, LoRA, QLoRA, BLIP	Large Language Models & RAG LangChain, LangGraph, AutoGen, LLaMA, RAG, GPT-4o, Azure OpenAI, ChromaDB, Pinecone, FAISS, Langfuse, Context Engineering	Computer Vision & Vision-Language Models OpenCV, CLIP, DINOv2, DINOv3, SAM
Machine Learning & Deep Learning PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, UNet	MLOps & Experimentation MLflow, SLURM, HPC	Cloud & DevOps AWS, Docker, Kubernetes, Terraform (IaC), GitHub Actions, Datadog
Programming & Data Analysis Python, SQL, Bash, NumPy, Pandas, Matplotlib, Seaborn	Software Development Tools FastAPI, ReactJS, Git, GitHub, Hugging Face Hub, Claude Code, Conda	

PROFESSIONAL EXPERIENCE

University of Siegen, AI Engineer 04/2026 – Present | Siegen, Germany

- Building PyTorch-based preprocessing and feature-extraction pipelines for multi-modal time-series and sensor data across 40+ subjects, standardizing inputs for downstream model training.
- Designing an automated labeling pipeline using vision-language models (CLIP, DINOv2, DINOv3) to annotate 100,000+ sensor windows, reducing manual labeling effort by 80% while reaching 90% classification accuracy.

Robert Bosch GmbH, AI Engineer 09/2025 – 03/2026 | Hildesheim, Germany

- Developed a controllable image-generation method adapting Stable Diffusion and FLUX to condition output on camera lens, sensor, viewpoint, and domain factors; the core contribution behind an ICPR 2026-accepted paper with Bosch Research and ETH Zürich.
- Assembled an automated BLIP-based captioning and metadata-filtering pipeline structuring 15+ object-detection datasets for generative fine-tuning, shrinking manual captioning time by 90%.
- Created a DINOv3-based automated evaluation framework to track generation quality and degradation across training iterations.
- Scaled training and evaluation workflows on a SLURM-based HPC cluster, reducing experiment iteration cycles by 40%.
- Validated generation outputs through a 150+ participant user study.

Robert Bosch GmbH, AI Engineer (Intern) 04/2025 – 09/2025 | Hildesheim, Germany

- Fine-tuned diffusion models with LoRA and QLoRA on domain-specific driving datasets to generate high-fidelity synthetic camera scenarios.
- Established reusable PyTorch/Diffusers/Transformers workflows to compare model variants, hyperparameters, and output quality.
- Containerized distributed training with Docker and tracked experiments in MLflow, trimming local setup time by 91% and enabling fully reproducible model tracking.
- Reduced release-to-validation time from 4 weeks to 1.5 weeks through automated dataset and experiment tracking.

Fraunhofer Institute for Mechatronic Systems Design, GenAI Engineer 06/2024 – 03/2025 | Paderborn, Germany

- Productionized a GPT-4o-powered document intelligence microservice via FastAPI, reaching 97% OCR parsing accuracy on handwritten historical and policy documents.
- Architected a production RAG engine (LangChain, ChromaDB, Pinecone) with hybrid search and metadata filtering, improving retrieval-driven compliance review speed by 40%.
- Delivered a ReactJS front end for document upload, retrieval, analysis, and AI-assisted review.

S&P Capital IQ Private Limited, Site Reliability Engineer 08/2022 – 09/2023 | Hyderabad, India

- Built and scaled AWS infrastructure with Terraform across 10+ services, sustaining 99.9%+ uptime.
- Engineered Datadog-based observability, cutting Mean Time to Detection by 70%.
- Automated cost, metric, and incident-reporting pipelines within a 7-member global infrastructure team.

Brane Services Private Limited, Project Intern 05/2022 – 07/2022 | Hyderabad, India

- Designed 20+ validation test cases and QA checklists, raising application test coverage by 80% across three release cycles.
- Tracked and validated 30+ application defects with a 5-member team, shortening release verification effort by 30%.

Advanced Data Processing Research Institute (ADRIN) 01/2022 – 06/2022 | Hyderabad, India
Machine Learning Engineer Intern

- Developed a TensorFlow-based voice-command recognition system for UAV/UGV control, reaching 95% real-time accuracy across 17 commands, deployed on Raspberry Pi at sub-10ms inference latency.

PROJECTS

German Legal AI Assistant & Enterprise RAG Engine

- Assembled a multilingual (German/English) RAG pipeline over 9,000+ metadata-tagged legal document chunks using MiniLM-L12 embeddings and citation-grounded response generation.
- Implemented semantic-hallucination monitoring with Langfuse and 40 automated tests, achieving sub-200ms latency on metadata-aware search.

AskDoc - Research Paper RAG Assistant

- Developed a document QA system converting research papers into searchable embeddings for natural-language querying, with source-grounded answer generation over FAISS vector search.

Jigsaw Puzzle Solver using Computer Vision

- Designed an end-to-end pipeline combining SAM segmentation, projective transformations, and graph-based matching to reconstruct large-scale jigsaw puzzles, reaching 80% solve accuracy on complex configurations.

Denoise Microscopic Data with Deep Learning

- Trained a UNet-based denoising model achieving ~54% PSNR improvement over classical filters (Gaussian, Median, Box) across two benchmark datasets.

AWARDS AND ACHIEVEMENTS

Deutschlandstipendium, Studienförderfonds Siegen e.V. 10/2024
Merit-based scholarship supporting academic excellence and commitment awarded to top ~1% of students in Germany

Best Project of 2021-22, Vasavi College of Engineering 07/2022
1st Prize for Best Final Year Project (~120 competing teams)

LANGUAGES

English | **German** — Conversational (B1) | **Hindi**

PUBLICATIONS

- S. Godavarthy et al., “MULTI: Disentangling Camera Lens, Sensor, View, and Domain for Novel Image Generation,” *International Conference on Pattern Recognition (ICPR)*, 2026. **Accepted for Oral Presentation.** [🔗](#)
- S. Godavarthy et al., “X-MULTI: VLM-based Imaging Factor Disentanglement for Factor-Aware Image Synthesis,” *2nd Workshop on Multimodal Large Language Models for Unified Comprehension and Generation (MUCG), European Conference on Computer Vision (ECCV)*, 2026. **Accepted Paper.**